Modeling COVID-19 Trends in the United States

Nathaniel E. Helwig

Associate Professor of Psychology and Statistics
University of Minnesota

September 28, 2020

Table of Contents

Table of Contents

Background and Motivation

Understanding COVID-19 trends in the US is important for…

- government policy makers
- health care providers
- public health researchers

Obtaining accurate estimates of COVID-19 trends in the US is a difficult task, due to the lack of a

1. national health care system
2. coordinated government response

Separating the Signal from the Noise

Estimates of COVID-19 trends must be constructed from noisy data.

- Each state's health department reports data differently
- No consensus on how to report the number of new cases

Noise in the case count data can occur for a variety of reasons:

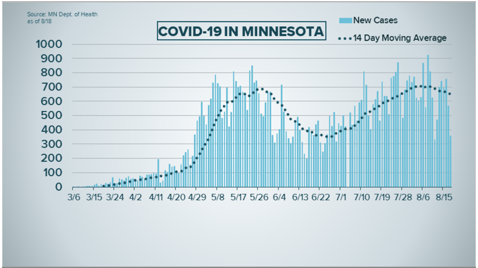
- data entry / reporting errors
- backlog of tests reported at once
- change in status of confirmed case

Need some method for separating the signal (i.e., trend) from the noise

Illustrations of Case Counts with Average Trends

Policy makers and researchers often rely on epidemic curves

- number of cases (y-axis) as a function of time (x-axis)
- Moving average estimates are often included to visualize trends



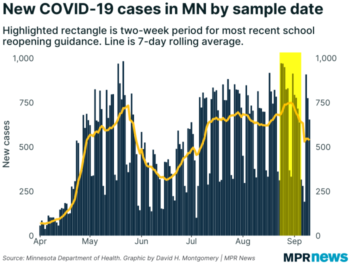


Figure 1: COVID-19 trends from kare11.com (left) and mprnews.org (right)

Table of Contents

Moving Average Estimates

The k-day moving average estimate at day t is defined as

$$\hat{\mu}_t = \begin{cases} \frac{y_{t-(k-1)/2} + \cdots + y_{t-1} + y_t + y_{t+1} + \cdots + y_{t+(k-1)/2}}{k} & \text{if } k \text{ is odd} \\ \frac{y_{t-k/2} + \cdots + y_{t-1} + y_{t+1} + \cdots + y_{t+k/2}}{k} & \text{if } k \text{ is even} \end{cases}$$

which is the average of the counts y_t within a k-day window around t.

- Could also use an asymmetric window: $\hat{\mu}_t = \frac{1}{k} \sum_{i=0}^{k-1} y_{t-i}$

Moving averages are simple, but not ideal for estimating latent trends:

- highly sensitive to outliers in the data
- no way to asses the uncertainty of the estimate
- cannot be used to forecast future case counts

Nonparametric Negative Binomial Regression

Nonparametric regression extends GLM (McCullagh and Nelder, 1989) to the situation where μ_t has some unknown form (Helwig, 2020).

The generalized nonparametric regression model assumes that

$$g(\mu_t) = \eta(t)$$

where the two terms are defined as

- $g(\cdot)$ is some known (user-specified) link function[1]
- $\eta(\cdot)$ is some unknown smooth function that is to be estimated

Goal: Find estimate $\hat{\eta}(t)$ that balances fitting and smoothing the data.

[1] In this application, the assumed link function is $\log(\mu_t) = \eta(t)$ for all t.

Table of Contents

Simulation Study Design

Simulated case count data for $n = 200$ days with different trends.

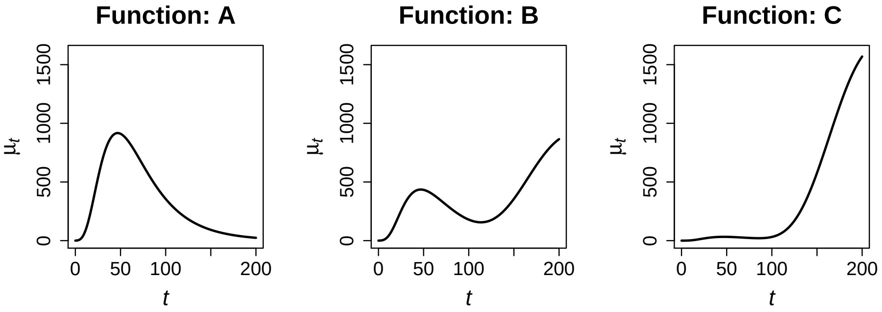


Figure 2: **Simulation design.** Data generating mean functions for simulation study. A) successful mitigation, B) attempted but unsuccessful mitigation, and C) no attempted mitigation.

Simulation Study Results

To compare trend estimates, use Relative RMSE $= \sqrt{\dfrac{\frac{1}{n}\sum_{t=1}^{n}(\hat{\mu}_t - \mu_t)^2}{\frac{1}{n}\sum_{t=1}^{n}\mu_t^2}}$

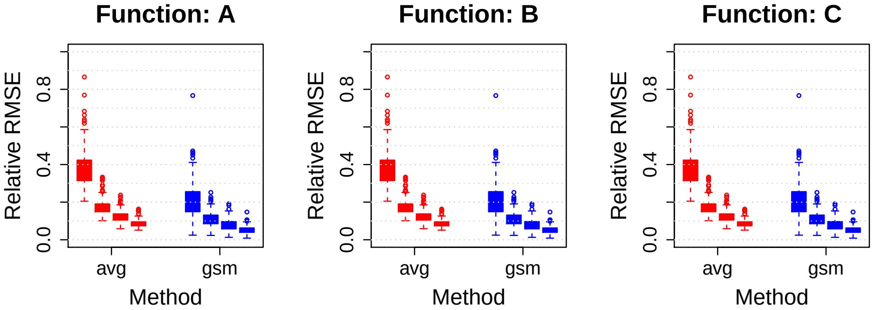


Figure 3: **Simulation results.** Relative RMSE using 7-day moving average (avg) and nonparametric negative binomial regression (gsm) for each data generating mean function. Within each subplot, the four boxes associated with each method denote the results for the four size parameters $\theta \in \{1, 5, 10, 20\}$.

Table of Contents

COVID-19 Data from The New York Times

Data used in this example are from The New York Times and are based on reports from various state and local health agencies.

- Available from The New York Times GitHub repository[2]

For each of the 50 states in the US, the dataset contains a daily time series of the cumulative COVID-19 case count.

Case count dates range from 2020-01-21 (first reported case in the US) to 2020-09-26 (day before the data were downloaded).

[2] `https://github.com/nytimes/covid-19-data`

Correcting Errors in the Case Count Data

Need to convert the cumulative case counts (provided in the dataset) into the daily number of new cases.

- x_t is the cumulative number of new cases on day t
- $y_t = x_t - x_{t-1}$ is the number of new cases on day t

Problem: this method produces some negative y_t due to errors in data

Solution: use isotonic regression (Barlow et al., 1972) to correct data

$$\min \sum_{t=1}^{n} (x_t - \hat{x}_t)^2 \quad \text{subject to } \hat{x}_t \leq \hat{x}_{t+1} \text{ for all } t$$

and then calculate the corrected difference scores $y_t = \hat{x}_t - \hat{x}_{t-1}$

Analysis of COVID-19 Case Count Trends

Compare moving average and generalized nonparametric regression

- moving average fit via `supsmu` function in R (R Core Team, 2020)
- nonparametric regression fit via **npreg** R package (Helwig, 2020)

Analyze data separately from each of the 50 states because

- makes it possible to compare moving average and NP regression
- makes it possible to compare results to local health agencies

Use nonparametric regression to forecast new case trends for one week

COVID-19 Case Trends for Minnesota

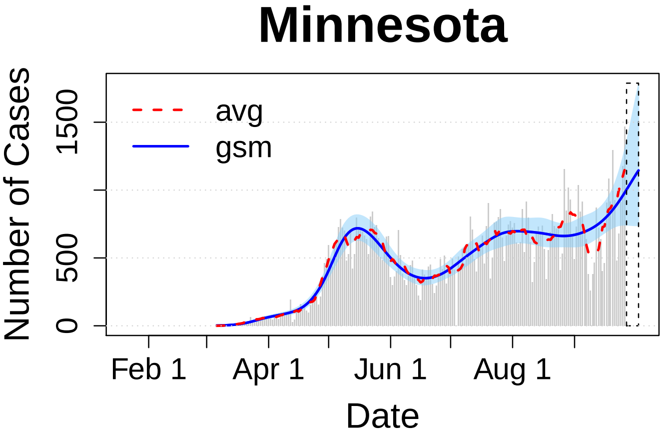


Figure 4: Daily COVID-19 new case count trends for Minnesota.

COVID-19 Case Trends for Wisconsin

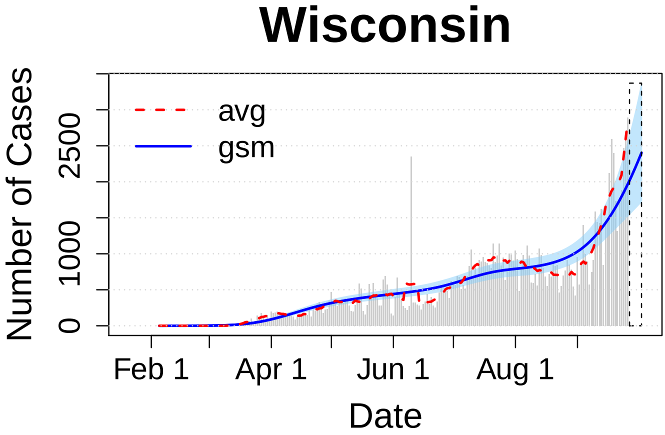


Figure 5: Daily COVID-19 new case count trends for Wisconsin.

COVID-19 Case Trends for Illinois

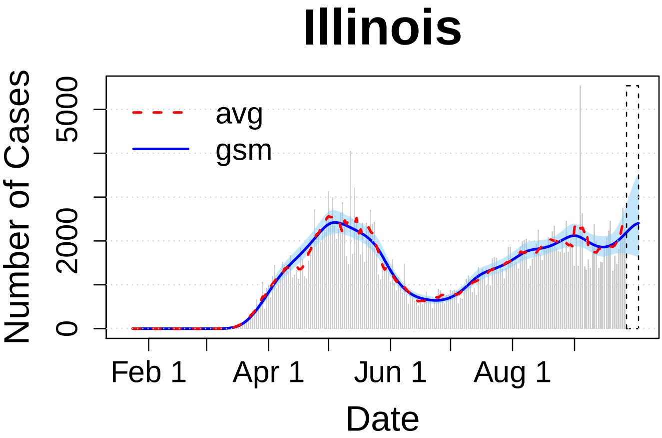


Figure 6: Daily COVID-19 new case count trends for Illinois.

COVID-19 Case Trends for New York

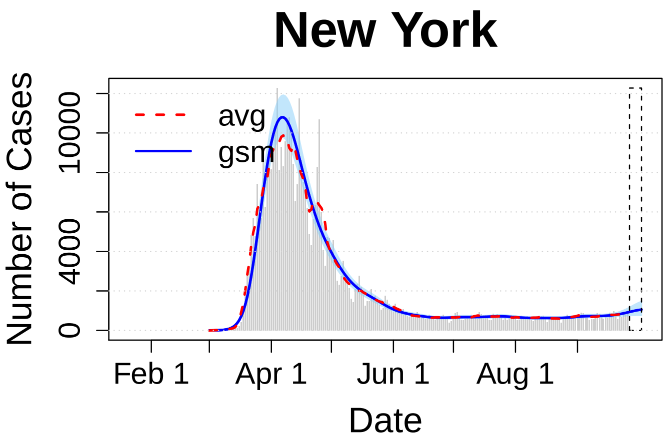


Figure 7: Daily COVID-19 new case count trends for New York.

COVID-19 New Case Trends for United States

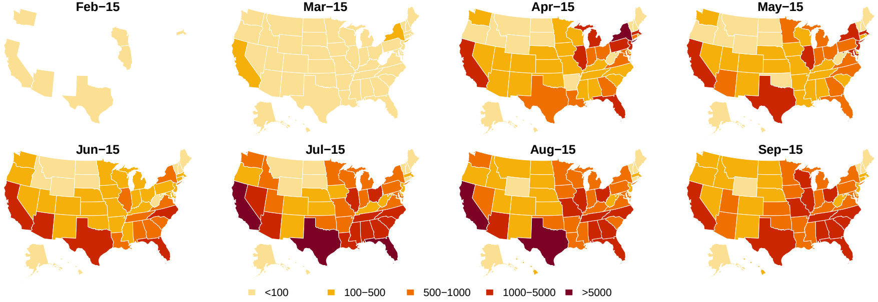


Figure 8: Monthly COVID-19 new case count trends for US.

Table of Contents

Findings and Recommendations

Nonparametric regression offers several benefits over moving average:

- less prone to capturing noise
- provides uncertainty estimates
- imputes missing data and forecasts future data

Can use NP regression for real-time trend estimation and forecasting

- Public policy decisions can be assessed and updated daily
- Decisions would be less prone to anomalous data points

COVID-19 trends are steadily rising in several states (including MN).

References

Barlow, R. E., D. J. Bartholomew, J. M. Bremner, and H. D. Brunk (1972). *Statistical Inference Under Order Restrictions: The Theory and Application of Isotonic Regression*. London: Wiley.

Helwig, N. E. (2020). Multiple and generalized nonparametric regression. In P. Atkinson, S. Delamont, A. Cernat, J. W. Sakshaug, and R. A. Williams (Eds.), *SAGE Research Methods Foundations*. SAGE.

Helwig, N. E. (2020). *npreg: Nonparametric Regression via Smoothing Splines*. R package version 1.0-4.

McCullagh, P. and J. A. Nelder (1989). *Generalized Linear Models* (Second ed.). London: Chapman and Hall.

R Core Team (2020). *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. R version 4.0.2.